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Digital-to-analog converter architecture for audio amplifiers

Abstract

In some embodiments, a digital-to-analog converter (DAC) architecture can include an array having a total number of bit cells, and a control system configured to activate a selected number of the total number of bit cells and to deactivate the remaining bit cells. The selected number can be variable, such that the array consumes a quiescent current that depends on the selected number. The control system can be further configured to change the selected number when a signal condition exceeds a threshold duration.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims priority to U.S. Provisional Application No. 63/250,188 filed Sep. 29, 2021, entitled TELESCOPING DAC ARCHITECTURE, the disclosure of each of which is hereby expressly incorporated by reference herein in its respective entirety.

BACKGROUND

Field

(1) The present disclosure relates to audio amplifier circuits for wearable audio devices such as earbuds or headphones.

Description of the Related Art

(2) A wearable audio device can be worn by a user to allow the user to enjoy listening of an audio content stream being played by a mobile device. Such an audio content stream may be provided from the mobile device to the wearable audio device through, for example, a short-range wireless link. Once received by the wearable audio device, the audio content stream can be processed by one or more circuits to generate an output that drives a speaker to generate sound waves representative of the audio content stream.

(3) It is desirable to have the foregoing sound waves provide reproduction of the audio content stream with high-fidelity. It is also desirable for the wearable audio device to operate in a power-efficient manner, since such devices are commonly powered by batteries having limited capacities.

SUMMARY

(4) In some implementations, the present disclosure relates to a digital-to-analog converter (DAC)

architecture that includes an array having a total number of bit cells, and a control system configured to activate a selected number of the total number of bit cells and to deactivate the remaining bit cells. The selected number is variable, such that the array consumes a quiescent current that depends on the selected number. The control system is further configured to change the selected number when a signal condition exceeds a threshold duration.

(5) In some embodiments, the DAC architecture can be configured to convert a digital signal stream into a respective analog audio signal stream.

(6) In some embodiments, some or all of the foregoing DAC architecture can be implemented on a semiconductor die, a packaged module, or some combination thereof.

(7) In some embodiments, some or all of the foregoing DAC architecture can be implemented in a wireless device such as a wireless headphone or a wireless earphone.

(8) For purposes of summarizing the disclosure, certain aspects, advantages and novel features of the inventions have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment of the invention. Thus, the invention may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a system that includes a wearable audio device in communication with a host device, where the wearable audio device includes an audio amplifier circuit.

(2) FIG. 2 shows that the wearable audio device of FIG. 1 can be implemented as a device configured to be worn at least partially in an ear canal of a user.

(3) FIG. 3 shows that the wearable audio device of FIG. 1 can be implemented as part of a headphone configured to be worn on the head of a user, such that the audio device is positioned on or over a corresponding ear of the user.

(4) FIG. 4 shows that in some embodiments, the audio amplifier circuit of FIG. 1 can include a number of functional blocks.

(5) FIG. 5 shows a block diagram of an audio amplifier circuit that is a more specific example of the audio amplifier circuit of FIG. 4.

(6) FIG. 6 shows a block diagram that includes a digital audio path block that is a more specific example of the digital audio path block of FIG. 5.

(7) FIG. 7 shows a block diagram of an amplifier block that is a more specific example of the amplifier block of FIG. 5.

(8) FIG. 8 shows a block diagram of a digital logic circuit block that is a more specific example of the digital logic circuit block of FIG. 5.

(9) FIG. 9 shows a gain compute block that is a more specific example of the gain compute block of FIG. 8.

(10) FIG. 10 shows various functional blocks of the audio amplifier circuit of FIG. 5 configured to provide a gain calibration sub-system.

(11) FIG. 11 shows a more detailed example of the gain calibration block of FIG. 10.

(12) FIG. 12A depicts a digital-to-analog converter (DAC) having N-bit resolution, where N is a positive integer.

(13) FIG. 12B depicts the DAC of FIG. 12A in a block form.

(14) FIG. 13A depicts a DAC that can be a more specific example of the DAC of FIG. 12A.

(15) FIG. 13B depicts the DAC of FIG. 13A in a block form.

(16) FIG. 14 shows that in some embodiments, a DAC can include a variable bit cell array size.

DETAILED DESCRIPTION OF SOME EMBODIMENTS

(17) The headings provided herein, if any, are for convenience only and do not necessarily affect the scope or meaning of the claimed invention.

(18) FIG. 1 depicts a system **1010** that includes a wearable audio device **1002** in communication with a host device **1008**. Such communication, depicted as **1007** in FIG. 1, can be supported by, for example, a wireless link such as a short-range wireless link in accordance with a common industry standard, a standard specific for the system **1010**, or some combination thereof. In some embodiments, the wireless link **1007** includes digital format of information being transferred from one device to the other (e.g., from the host device **1008** to the wearable audio device **1002**).

(19) In FIG. 1, the wearable device **1002** is shown to include an audio amplifier circuit **1000** that provides an electrical audio signal to a speaker **1004** based on a digital signal received from the host device **1008**. Such an electrical audio signal can drive the speaker **1004** and generate sound representative of a content provided in the digital signal, for a user wearing the wearable device **1002**.

(20) In FIG. 1, the wearable device **1002** is a wireless device; and thus typically includes its own power supply **1006** including a battery. Such a power supply can be configured to provide electrical power for the audio device **1002**, including power for operation of the audio amplifier circuit **1000**. It is noted that since many wearable audio devices have small sizes for user-convenience, such small sizes places constraints on power capacity provided by batteries within the wearable audio devices.

(21) In some embodiments, the host device **1008** can be a portable wireless device such as, for example, a smartphone, a tablet, an audio player, etc. It will be understood that such a portable wireless device may or may not include phone functionality such as cellular functionality. In such an example context of a portable wireless device being a host device, FIGS. 2 and 3 show more specific examples of wearable audio devices **1002** of FIG. 1.

(22) For example, FIG. 2 shows that the wearable audio device **1002** of FIG. 1 can be implemented as a device (**1002a** or **1002b**) configured to be worn at least partially in an ear canal of a user. Such a device, commonly referred to as an earbud, is typically desirable for the user due to compact size and light weight.

(23) In the example of FIG. 2, a pair of earbuds (**1002a** and **1002b**) can be provided—one for each of the two ears of the user—and each earbud can include its own components (e.g., audio amplifier circuit, speaker and power supply) described above in reference to FIG. 1. In some embodiments, such a pair of earbuds can be operated to provide, for example, stereo functionality for left (L) and right (R) ears.

(24) In another example, FIG. 3 shows that the wearable audio device **1002** of FIG. 1 can be implemented as part of a headphone **1003** configured to be worn on the head of a user, such that the audio device (**1002a** or **1002b**) is positioned on or over a corresponding ear of the user. Such a headphone is typically desirable for the user due to audio performance.

(25) In the example of FIG. 3, a pair of audio devices (**1002a** and **1002b**) can be provided—one for each of the two ears of the user. In some embodiments, each audio device (**1002a** or **1002b**) can include its own components (e.g., audio amplifier circuit, speaker and power supply) described above in reference to FIG. 1. In some embodiments, one audio device (**1002a** or **1002b**) can include an audio amplifier circuit that provides outputs for the speakers of both audio devices. In some embodiments, the pair of audio devices **1002a**, **1002b** of the headphone **1003** can be operated to provide, for example, stereo functionality for left (L) and right (R) ears.

(26) FIG. 4 shows that in some embodiments, the audio amplifier circuit **1000** of FIG. 1 can include a number of functional blocks. More particularly, in FIG. 4, an audio amplifier circuit **1000** is shown to include a digital logic circuit block **1020**, an amplifier block **1022**, a power management block **1024**, and an ancillary block **1026**. Examples related to such blocks are described herein in greater detail.

(27) In FIG. 4, the audio amplifier circuit **1000** is shown to further include various interfaces to allow the audio amplifier circuit **1000** to interact with other devices external to the audio amplifier circuit **1000**. More particularly, an interface indicated as **1030** can be configured to support input/output (I/O) functionality with respect to a host device (e.g., **1008** in FIG. 1). An interface indicated as **1034** can be configured to support providing of electrical audio signals to a speaker (e.g., **1004** in FIG. 1). An interface indicated as **1032** can be configured to support providing of electrical power to various parts of the audio amplifier circuit **1000**. One or more ground pins collectively indicated as **1036** (GND) can be configured to provide a grounding connection for the audio amplifier circuit **1000** relative to, for example, the audio device **1002** of FIG. 1.

(28) FIG. 5 shows a block diagram of an audio amplifier circuit **1000** that is a more specific example of the audio amplifier circuit **1000** of FIG. 4. In FIG. 5, a digital logic circuit block, generally indicated as **1020**, can include a number of more specific functional blocks; an amplifier block, generally indicated as **1022**, can include a number of more specific functional blocks; a power management block, generally indicated as **1024**, can include a number of more specific functional blocks; and an ancillary block, generally indicated as **1026**, can include a number of more specific functional blocks. Similarly, an interface indicated as **1030** can include a number of pins to support input/output (I/O) functionality with respect to a host device; an interface indicated as **1034** can include a number of pins to support providing of electrical audio signals to a speaker; an interface indicated as **1032** can include a number of pins to support providing of electrical power to various parts of the audio amplifier circuit **1000**; and one or more ground pins collectively indicated as **1036** (GND) can be implemented to provide a grounding connection for the audio amplifier circuit **1000**.

(29) Referring to the example of FIG. 5, the digital logic circuit block **1020** can include a receiver (Rx) block **1040** configured to receive, for example, a pulse-density modulation (PDM) signal through a DATA pin of the interface **1030**. The PDM Rx block **1040** is shown to also receive a clock signal through a CLK pin of the interface **1030**. The PDM Rx block **1040** is shown to provide an output based on the input PDM signal.

(30) It will be understood that while various examples are described herein in the context of pulse-density modulation of signals, one or more features of the present disclosure can also be implemented utilizing other types of modulations including other types of pulse modulations.

(31) In FIG. 5, the digital logic circuit block **1020** can further include a digital audio path block **1042**. Such a block is shown to receive the output of the PDM Rx block **1040** and route the received signal to the amplifier block **1022**. Additional examples related to the digital audio path block **1042** are described herein in greater detail.

(32) As shown in FIG. 5, the digital logic circuit block **1020** can also include various blocks for providing control and calibration functionalities. For example, amplifier controller **1090**, resistance network control block **1064**, amplifier operating mode (e.g., HOR/ZOR mode) control block **1062**, inter-integrated circuit (I2C) auxiliary block **1092**, registers block **1094**, PDM detect block **1060** and loudness protection block **1066** can provide and/or support various control functionalities described herein. In another example, current ratio measurement calibration block **1068** and gain calibration block **1070** can provide calibration functionalities described herein. More particularly, the current ratio measurement calibration block **1068** can support generation of a reference signal for a loop circuit as described herein, and the gain calibration block **1070** can provide various functionalities for gain calibration as described herein.

(33) Referring to the example of FIG. 5, the amplifier block **1022** is shown to include a pulse-width modulation (PWM) controller **1050** configured to receive a feedforward digital signal from the digital audio path block **1042** of the digital logic circuit block **1020** (through a path indicated as **1043**) and generate control signals for an H-bridge driver **1052**. The H-bridge driver **1052** provides analog electrical audio signals HPN, HPP as outputs. Such electrical audio signals can be provided to a speaker through respective pins of the interface **1034**.

(34) In the example of FIG. 5, the amplifier block **1022** is configured as a digital PWM Class D amplifier. In addition to the H-bridge driver **1052** being pulse-width modulated by the PWM controller **1050** based on the feedforward digital signal from the digital audio path block **1042**, a closed-loop architecture is provided. Such a closed loop is shown to include an input resistance network **1080** coupled to the HPN and HPP outputs of the H-bridge driver **1052**, with the input resistance network **1080** being coupled to a loop filter **1046** through summing nodes **1081**, **1083**. An analog output from the loop filter **1046** is shown to be converted into a digital signal by an analog-to-digital converter (ADC) **1048** such as a successive approximation register (SAR) ADC. The digital signal from the SAR ADC **1048** is provided to the PWM controller **1050**.

(35) In the example of FIG. 5, the amplifier block **1022** is configured to provide a reference analog signal for the foregoing closed-loop circuit. More particularly, a digital signal from the digital audio path block **1042** is shown to be provided to a digital-to-analog converter (DAC) **1044** (through a path indicated as **1045**), and the resulting analog signal is provided to the summing nodes **1081**, **1083**. The summing nodes **1081**, **1083** are also shown to be provided with respective signals from a common-mode-limit (CML) amplifier **1082**.

(36) In the example of FIG. 5, the audio amplifier circuit **1000** is shown to include a gain calibration feature. Such a feature is shown to include a calibration ADC **1084** coupled to the HPN and HPP outputs of the H-bridge driver **1052** to provide a digital signal representative of the analog output signals of the H-bridge driver **1052**. The digital signal from the calibration ADC **1084** is shown to be provided to the gain calibration block **1070** of the digital logic circuit block **1020**.

(37) In the example of FIG. 5, the H-bridge driver **1052** shown to be provided with multiple levels of supply voltages (e.g., VBAT, VDD_A, VDD_B, VDD_D, VDD_E). Such multiple voltage levels can allow the H-bridge driver **1052** to operate with improved power efficiency.

(38) Additional examples concerning the amplifier block **1022** are described herein in greater detail.

(39) In FIG. 5, the power management block **1024** can include a number of functional blocks configured to provide and/or support providing of power to various parts of the audio amplifier circuit **1000**. For example, the power management block **1024** can be configured to provide routing of multiple supply voltage levels (e.g., VBAT, VDD_A, VDD_B, VDD_D, VDD_E) to the H-bridge driver **1052** of the amplifier block **1022**. For the example supply voltage levels, VBAT>VDD_A>VDD_B>VDD_D>VDD_E.

(40) Such supply voltages can be provided from source(s) external to the audio amplifier circuit **1000**, from internal source(s), or some combination thereof. In the example of FIG. 5, supply voltages VBAT, VDD_A and VDD_B are provided from external source(s); VDD_D may be provided from an external source or from an internal source implemented as a low drop out (LDO) regulator **1130**; and VDD_E is provided from an internal source implemented as a low voltage monitor (LVM) supply **1132**.

(41) Some or all of the foregoing voltages can be monitored by one or more voltage monitors. For example, a supply voltage monitor (SVM) **1120** is shown to monitor the voltages VBAT, VDD_A, VDD_B and VDD_D. Such an SVM can include low power low resolution ADCs that monitor the supply voltages and produce respective digital outputs representative of the supply voltages; and such monitored digital outputs can be utilized by other digital circuitry to control various functionalities of the audio amplifier circuit **1000**. In another example, the voltage VDD_E is shown to be self-monitored by the LVM supply **1132**.

(42) Referring to the example of FIG. 5, the power management block **1024** is shown to further include a reference (Ref) block **1110**. Such a reference block can be implemented as a low voltage, low power bandgap reference circuit configured to operate with a supply voltage (e.g., VDD_B) to produce a low reference voltage as an output. Such a reference voltage can be utilized for operation of an analog LDO regulator **1112** and a digital LDO regulator **1114**, as well as other functional blocks of the audio amplifier circuit **1000**. The analog LDO regulator **1112** can be implemented as

a lower power linear regulator configured to provide a desired voltage for a number of circuits of the audio amplifier circuit **1000**. The digital LDO regulator **1114** can be implemented as a low power linear regulator configured to provide a desired voltage for various digital logic and digital core circuits of the audio amplifier circuit **1000**.

(43) Referring to the example of FIG. 5, the power management block **1024** is shown to further include a low power oscillator (LPO) **1118**. Such an LPO can be configured to support operation of the audio amplifier circuit **1000**. The power management block **1024** is shown to further include a sensor block **1116** such as a temperature sensor. Such a sensor can be configured to detect operating condition(s) (e.g., temperature) of some or all of the audio amplifier circuit **1000**; and such sensed condition(s) can be utilized to support one or more functionalities (e.g., fault protection) for the audio amplifier circuit **1000**.

(44) In FIG. 5, the ancillary block **1026** is shown to include a power-on-reset (POR) block **1100**. Such a POR block can be implemented to provide a number of functionalities. For example, power-on reset functionality can be provided by the POR block **1100**, where the POR block **1100** monitors the RESET_B pin and supply voltage conditions to control and/or support power-on sequencing of various regulators, clock system and wall level shifters utilizing respective control signals (Pups). Once such power-on sequencing is achieved and the controlled components are operating in a stable manner, a release signal (Dig_reset_B) is provided to allow operation of various digital blocks.

(45) The POR block **1100** can also control and/or support a power-down sequence. Such a power-down sequence can be achieved in response to a control signal from a host device (e.g., setting RESET_B to a initiate power-down), or based on detection of one or more conditions. Such conditions can include, for example, a brownout detection and various fault detections.

(46) In FIG. 5, the ancillary block **1026** is shown to include a one-time programmable memory (OTP) **1102** and a block **1104** providing control and register functionalities for the OTP block **1102**. Such functionalities can include issuing of a control signal (I2C address) to the **120** block **1092** to load appropriate registers of the Registers block **1094** during a boot process.

(47) FIG. 6 shows a block diagram that includes a digital audio path block **1042** that is a more specific example of the digital audio path block **1042** of FIG. 5. In FIG. 6, a pulse-density modulation (PDM) receiver (Rx) block **1040** is shown to receive a PDM signal DATA and a clock signal CLK, and provide a PDM digital signal to the digital audio path block **1042**.

(48) More particularly, the PDM digital signal from the PDM Rx block **1040** is shown to be provided to a digital low-pass filter (PDM LPF) **1140**. Such a filter block can be configured to, for example, attenuate out-of-band noise in the received PDM digital signal (e.g., noise resulting from a transmit (Tx) modulation in a host device). The PDM LPF block **1140** of FIG. 6 can also be configured to convert the PDM input signal into an output digital signal having pulse-code modulation (PCM).

(49) In FIG. 6, the filtered PCM signal from the PDM LPF block **1140** is shown to be provided to an equalizer (EQ) block **1142**. Such an EQ block can be configured to support gain and mute functionalities, as well as high-output resistance (HOR) and/or zero-output resistance (ZOR) operating modes. The EQ block **1142** provides an output PCM signal to a calibration tone mixer **1144**. Additional details concerning the EQ block **1142** are provided herein.

(50) Referring to FIG. 6, the calibration tone mixer **1144** is shown to mix a calibration tone signal from a gain calibration block **1070** with the output PCM signal from the EQ block **1142**. Such mixing functionality can be provided during a calibration process such as an HOR/ZOR gain calibration process. Additional details concerning the HOR/ZOR gain calibration process are provided herein.

(51) In FIG. 6, the output of the calibration tone mixer **1144** is also shown to be processed through a number of blocks before being provided a digital-to-analog converter (DAC) **1044** (also **1044** in FIG. 5) through a path **1045** for a closed-loop circuit as described herein. More particularly, the

output (PCM signal) of the calibration tone mixer **1144** is shown to be provided to a signal limiter block **1146**. Such a signal limiter can be configured to maintain a threshold limit for a DAC modulator (e.g., delta sigma modulator (DSM)) **1148** and also improve handling of low supply voltage operating conditions (e.g., when a HOR mode load reactance causes supply voltage requirements of an H-bridge drive to increase beyond a pure resistive load condition).

(52) Referring to FIG. 6, it is noted that for the foregoing low voltage operating conditions, a SAR ADC (e.g., **1048** in FIG. 5) in the closed-loop circuit can be driven into saturation if the power supply is insufficient to provide the required peak voltage for the H-bridge driver (**1052** in FIG. 5) needed to develop an output voltage at the load. If such a SAR ADC is saturated, the corresponding loop filter (**1046** in FIG. 5) is saturated and is slow to recover from such overload conditions, resulting in undesirable audio artifacts. Thus, it is desirable to have the SAR ADC and loop filter prevented from entering saturation. Such saturation-prevention can be accomplished by limiting the digital audio signal in digital path of the closed-loop circuit to prevent the closed-loop circuit from trying to generate an unachievable output voltage and thereby push the SAR ADC and loop filter into saturation.

(53) Referring to FIG. 6, the signal limiter **1146** can be configured to prevent the SAR ADC and loop filter from being in saturation due to insufficient supply voltage while trying to produce an output voltage from the corresponding digital input signal. The signal limiter **1146** can limit (e.g., by clipping the digital audio signal to a clip level) based on either or both of loop filter saturation detection signal (LF Sar. Det.) and SAR ADC output level (SAR ADC out) to keep the SAR ADC and the loop filter from overloading. It is noted that the SAR ADC output level can be monitored to have the signal limiter to clip the digital audio signal when the SAR ADC output level is close to saturation.

(54) In FIG. 6, the signal limiter block **1146** is shown to provide an output signal to a DAC modulator (e.g., delta sigma modulator (DSM)) **1148**. As described above, such an output of the signal limiter block **1146** can be clipped to avoid the SAR ADC and loop filter from being in saturation. The output of the signal limiter block **1146** can also prevent the DAC DSM **1148** from overloading.

(55) In the example of FIG. 6, the DAC DSM block **1148** can be configured to re-modulate a higher-bit input signal (e.g., 24-bit signal), through delta-sigma modulation, into a lower-bit signal (e.g., 9-bit signal) that is appropriate for a dynamic element matching (DEM) block **1150**. Accordingly, the example 9-bit DSM output signal drives the DEM block **1150** which can be implemented as a digital block configured to, for example, randomize a pattern of 512-bit cell drive to the DAC **1044** in a manner to linearize the DAC's response for use as a multi-bit delta-sigma DAC. Such a configuration can provide a desirable reference audio signal for the closed-loop circuit described herein.

(56) In FIG. 6, a feedforward signal is shown to be provided to a PWM controller (**1050** in FIG. 5) from an output of the signal limiter **1146** through a path **1043**. As described herein, such a feedforward signal may or may not include a calibration tone signal mixed therein, depending on operating status of the gain calibration process.

(57) In addition to the PDM LPF block **1140**, EQ block **1142**, mixer **1144**, signal limiter block **1146**, DSM block **1148** and DEM block **1150** that can be generally referred to as the digital audio path block **1042**, FIG. 6 also shows a number of blocks that support various functionalities of the audio amplifier circuit (**1000** in FIG. 5). For example, a PDM detection block **1060** (also **1060** in FIG. 5) is shown to be coupled to the PDM Rx block **1040**. The PDM detection block **1060** can be configured to detect one or more states of PDM digital audio interface, including one or more fault conditions, to support operation and control of the audio amplifier circuit **1000**.

(58) In another example, a loudness protection block **1066** (also **1066** in FIG. 5) is shown to be coupled to the PDM Rx block **1040**. The loudness protection block **1066** can be configured to monitor both of two output channels (main channel and auxiliary channel) of the PDM Rx block

1040. Upon loudness detection (e.g., digital audio signal exceeding a threshold), the loudness protection block **1066** can issue a fault signal to provide a muting or fault condition functionality.

(59) In FIG. **6**, the loudness protection block **1066** can include a pair of filter stages for the two output channels of the PDM Rx block **1040**. Each filter stage can include a cascaded low-pass filter and high-pass filter structure configured to approximate a frequency response such as an A-weighted frequency response. The filtered outputs can be provided to respective absolute value circuits, and outputs thereof can then be provided to comparator circuits and compared against a programmed threshold for each of the main channel, auxiliary channel and main-minus-auxiliary values. Logic outputs of the comparators can be sent to a multiplexer that can trigger a fault signal depending on a combination of the three comparator outputs.

(60) In FIG. **6**, a number of functional blocks are shown to be coupled to and/or be related to one or more functional blocks of the digital audio path block **1042**. For example, a digital-to-analog converter (DAC) **1044** is shown to be coupled to an output of the DEM block **1150**. Such a DAC can be utilized to provide a reference signal for a closed-loop architecture of the audio amplifier circuit (**1000** in FIG. **5**). Additional details concerning the closed-loop architecture are described herein.

(61) In another example, a HOR/ZOR state control block **1062** is shown to be coupled to the EQ block **1142**. Such a control block, along with a resistance control (Rout ctrl) block **1064** and a resistance network **1080**, can be utilized to provide various functionalities associated with high-output resistance (HOR) and zero-output resistance (ZOR) operating modes. Additional details concerning such operating modes are described herein.

(62) In yet another example, a gain calibration block **1070** is shown to provide a calibration tone to the mixer **1144** based on inputs from signal limiter **1146** and a calibration ADC **1084**. Additional details concerning gain calibration of the audio amplifier circuit (**1000** in FIG. **5**) are described herein.

(63) In FIG. **6**, operations of various functional blocks are shown to be supported by audio path registers **1094**. Such registers can be a part of or associated with the registers block **1094** of FIG. **5** and be configured in a similar manner.

(64) FIG. **7** shows a block diagram of an amplifier block **1022** that is a more specific example of the amplifier block **1022** of FIG. **5**. The amplifier block **1022** of FIG. **7** includes a digital PWM synthesis class D amplifier architecture. It is noted that unlike a purely analog class D amplifier architecture where pulse-width modulation (PWM) is analog, the amplifier architecture of the amplifier block **1022** includes pulse width modulation of H-bridge drivers being developed via digital pulse-width modulation by a digital PWM controller **1050**. The amplifier architecture of the amplifier block **1022** also includes a closed-loop control feature having a high loop gain error amplifier and an ADC digitizer.

(65) As described in reference to FIG. **6**, a feedforward digital signal is provided to the PWM controller **1050** from the digital audio path **1042** of the digital logic circuit **1020**. More particularly, the feedforward digital signal is provided to the PWM controller **1050** from an output of the signal limiter block **1146**, through a signal path **1043**. In FIG. **7**, such a signal path is also indicated as **1043**.

(66) As also described in reference to FIG. **6**, the digital signal from the output of the signal limiter block **1146** is also provided to the DAC **1044** through the DSM block **1148** and the DEM block **1150**. An analog signal from the output of the DAC **1044** is utilized as a reference audio signal for the above-referenced closed-loop of the amplifier block **1022** of FIG. **7**.

(67) Referring to FIG. **7**, it is noted that the feedforward digital audio signal that is provided to the PWM controller **1050** (through the path **1043**) is utilized to create most of a signal that determines pulse modulation for the H-bridge driver **1052**. More particularly, the PWM controller **1050** is shown to include a pulse generator **1166** that generates control signals HPP_ctrl, HOR_ctrl, HPN_ctrl based mostly on the feedforward digital audio signal provided through the path **1043** and

a mixer **1164**. The control signal HPP_ctrl is provided to a ZOR HPP driver **1170** to generate an analog audio signal HPP when in a ZOR mode; the control signal HOR_ctrl is provided to an HOR driver **1172** to generate an analog audio signal HPP when in an HOR mode; and the control signal HPN_ctrl is provided to a ZOR/HOR HPN driver **1174** to generate an analog audio signal HPN when in either of the ZOR and HOR modes. The analog signals HPP and HPN are shown to drive a speaker **1004** to generate sound.

(68) Referring to FIG. 7, it is also noted that an error signal generated by the closed-loop is utilized to develop the remainder of the signal that determines pulse modulation for the H-bridge driver **1052**. Such an error signal is shown to be provided to the pulse generator **1166** from a successive approximation register (SAR) ADC **1048** through a digital loop filter (DLF) **1162** and the mixer **1164**. Such an error signal resulting from the closed-loop provides improved audio performance of the amplifier circuit. Examples related to such a closed-loop are described herein in greater detail.

(69) In the example of FIG. 7, the H-bridge driver **1052** is shown to include a ZOR HPP driver **1170**, an HOR driver **1172** and a ZOR/HOR HPN driver **1174**, and a resistance network **1080** (also **1080** in FIG. 6) is shown to include a sense resistance (R_s), HOR feedback resistances (R_{h1} , R_{h2}) and ZOR feedback resistances (R_{z_p} , R_{z_n}). With such drivers, a ZOR mode can be implemented so that the ZOR HPP (**1170**) and ZOR/HOR HPN (**1174**) drivers directly drive the speaker **1004** (with signals through HPP and HPN nodes), and the ZOR feedback resistances (R_{z_p} , R_{z_n}) of the resistance network **1080** directly sense the voltage across the speaker load (HPP and HPN).

Accordingly, the ZOR HPP (**1170**) and ZOR/HOR HPN (**1174**) drivers are directly connected to the HPP and HPN nodes, and thus the speaker load, during the ZOR mode.

(70) Referring to FIG. 7, an HOR mode can be implemented so that the HOR feedback resistances (R_{h1} , R_{h2}) sense a voltage across the sense resistance R_s (e.g., an on-chip current sense resistor), where R_s can be adjusted to be same or close to the resistance (R_L) of the speaker load. With such a configuration, the closed-loop operation can force the voltage signal across the sense resistance R_s to be representative of the input signal provided to the H-bridge driver **1052**. Accordingly, the resulting current through the speaker load causes an output voltage of the H-bridge driver **1052** to be equal to or representative of the input signal provided to the H-bridge driver **1052** times the speaker impedance.

(71) It is noted that during the foregoing HOR mode operation, the ZOR HPP driver **1170** is turned off. It is also noted that the resistance R_s is in series with the HOR driver **1172** and the HPP node. Accordingly, the speaker load is driven through the high impedance of the HPP node. The ZOR/HOR HPN driver **1174** drives the other side of the speaker load.

(72) It is noted that the foregoing HOR mode can be utilized to address a low-level electromagnetic-coupled noise problem. For example, the high output resistance mode can attenuate the noise at the speaker load. More particularly, in the HOR mode, the speaker load is driven in a high-output-impedance mode as a current source mode output instead of a voltage source mode output. Accordingly, the H-bridge driver **1052** forces a high-fidelity audio current waveform into the speaker load, regardless of load impedance, nonlinearities and/or noise injections.

(73) The foregoing HOR mode can be calibrated an adjustment of the sense resistance R_s and a digital HOR calibration gain factor applied in one or more calibration blocks. In some embodiments, such gain calibration can be achieved periodically to equal the gain in the ZOR mode. Examples related to such gain calibration are described herein in greater detail.

(74) In the example of FIG. 7, the PWM amplifier **1022** utilizes modulation frequency and supply voltage scheme to provide high performance and efficient operation of the H-bridge driver **1052**. The modulation frequency can have a value of several MHz for a pulse width update rate to provide an update period. Such an update period is divided into N ticks utilizing a clock signal. An output pulse width can range from 1 to M times the tick width. Accordingly, the output pulse width can have a minimum value of $1 \times (\text{tick width})$ and a maximum value of $M \times (\text{tick width})$.

(75) Referring to FIG. 7, the foregoing supply voltage scheme can include utilization of multiple supply voltages for the output pulses. For example, voltages $V_{BAT} > V_{DD_A} > V_{DD_B} > V_{DD_D} > V_{DD_E}$ can be provided to an H-bridge driver supply circuit **1160** for the output voltages. Such multiple voltages provided to the output pulses can provide improved efficiency during operations at different signal levels. For example, for lower level signals, lower voltages can be utilized; and for higher level signals, higher voltages can be utilized. To achieve such functionality, the H-bridge driver **1052** can include multiple driver transistors configured to allow dynamic switching of voltages to any of the multiple values based on encoded control signals from the PWM controller **1050**.

(76) For example, and referring to FIG. 7, the ZOR HPP and ZOR/HOR HPN drivers (**1170**, **1174**) are utilized for ZOR mode, and the HOR and ZOR/HOR HPN drivers (**1172**, **1174**) are utilized for HOR mode. Depending on the mode of operation (ZOR or HOR), amplitude of input signal and PWM encoding rules, the PWM controller **1050** connects the respective drivers to one of the available supplies provided through the H-bridge driver supply circuit **1160** to create a zero, positive or negative differential drive across the output load nodes HPP and HPN.

(77) As described above, the PWM amplifier **1022** utilizes modulation frequency such that a pulse width update is provided during a corresponding period. Thus, the PWM controller **1050** can select the supply voltages and pulse width for the respective drivers. For example, the PWM controller **1050** can select the largest pulse width and lowest supply possible during each update period. Such selections of pulse width and supply voltage level can result in the lowest or reduced PWM quantization error and best or improved power efficiency.

(78) Configured in the foregoing manner, the output of the H-bridge driver **1052** can be taken from the differential voltage on the HPP and HPN nodes and directly drive the speaker **1004**. It is noted that such an output differs from traditional class D amplifiers in that the foregoing output appears as high frequency, multi-voltage-level stepping/switching activity. Such voltage stepping activity is a notable property of the architecture of the PWM amplifier **1022**. For the ZOR mode of operation and for audio signals such as sine waves, the voltage level stepping can follow the envelope of the audio signal. In the HOR mode of operation, the HPP and HPN single-ended output switching appears different than in the ZOR mode. The behavior of the signal on the HPP and HPN nodes can depend on the polarity of the output signal. When measuring the HPP and HPN nodes single-ended to ground, during parts of the output signal cycle, the HPP and HPN waveforms may not resemble the audio envelope in the same way as in the ZOR mode. Such a difference can result from the HOR mode's selection of HPP node voltage that forces the PWM controller **1050** to produce switching behavior that holds the HPP node high. Therefore, the HPP node is held high for a significant part of the waveform cycle while the HPN node is switching.

(79) In the PWM amplifier **1022** of FIG. 7, the closed-loop can be configured and operated as follows. As described herein, such a closed-loop can provide an error signal that is utilized for development of a signal that determines pulse modulation for the H-bridge driver **1052**, thereby providing improved audio performance of the amplifier circuit.

(80) Referring to FIG. 7, the outputs HPP and HPN of the H-bridge driver **1052** can be fed back to the summing nodes **1081**, **1083** through the resistance network **1080**. More particularly, the output node HPP is shown to be coupled to the summing node **1081** through a resistance R_{z_p} and a respective mixer also being provided with a ZOR mode signal as an input. Thus, the output of the mixer is shown to be added with a respective reference output of the DAC **1044** at the summing node **1081**; and the summed signal is shown to be provided to the loop filter **1046**. Similarly, the output node HPN is shown to be coupled to the summing node **1083** through a resistance R_{z_n} and a respective mixer also being provided with a ZOR mode signal as an input, such that the output of the mixer is shown to be added with a respective reference output of the DAC **1044** at the summing node **1083**; and the summed signal is shown to be provided to the loop filter **1046**.

(81) Configured in the foregoing manner, the loop filter **1046** is provided with a signal

representative of a differential error between the outputs (HPP, HPN) of the H-bridge driver **1052** and the reference signal provided by the DAC **1044**. In the example of FIG. 7, the each of the summing nodes **1081**, **1083** is shown to be provided with a signal from the common-mode limiting (CML) amplifier **1082**. Such signals from the CML amplifier **1082** can be utilized to limit the input common-mode voltage of the loop filter **1046**.

(82) In the example of FIG. 7, the loop filter **1046** can include a 5th order high gain loop filter to provide an output to a low-power, high-speed SAR ADC **1048** to digitize the loop filter output. The digitized output of the SAR ADC **1048** is shown to be provided to the PWM controller **1050**, where it is utilized by the PWM controller **1050** along with the feedforward signal (provided through the path **1043**) to generate PWM control signals for the H-bridge driver **1052**.

(83) Referring to FIG. 7, it is noted that in the foregoing closed-loop, without any compensation, an inductance of the speaker's driver element can lead to significant differences in the open-loop frequency response at high frequencies (e.g., greater than 100 KHz) between the HOR and ZOR modes. Such an effect is due to the current-mode drive of the sense resistance working into the frequency dependent impedance of the speaker inductance during the HOR mode of operation.

(84) In the closed-loop circuit of FIG. 7, the loop includes the loop filter **1046**, SAR ADC **1048**, PWM controller **1050**, H-bridge driver **1052** and resistance network **1080**. In addition to such components, a digital loop filter (DLF) **1162** can be provided to provide compensation for loop stability in situations where processing delays are present. In some embodiments, such a DLF can be configured to provide compensation by insertion of a programmable digital filter that includes parallel arrangement of finite impulse response (FIR) and infinite impulse response (IIR) sections, between the SAR ADC **1048** and the normal input location (mixer **1164** in FIG. 7). Such a DLF can be configured to provide a response that includes phase compensation in response to the effect of the inductor in the HOR mode, as well as shaping of one or more characteristics of the closed-loop.

(85) As described herein, when the PWM amplifier **1022** of FIG. 7 is in ZOR mode, the gain is determined by the resistances of the resistance network **1080** working against the DAC (**1044**) output current. The feedback resistances (R_{z_p} , R_{z_n}) sense the voltage across the HPP and HPN nodes, and provides feedback to the loop filter inputs. The closed-loop with such a feedback can force the output of the PWM amplifier **1022** to be adjusted to equal or approximately equal the digital input with a net gain (e.g., $G=1$ such that a 0 dBFS input produces a 0 dBv output).

(86) In HOR mode, however, the gain is determined differently, since the HOR mode utilizes a current-mode output where the signal current is produced across the sense resistance (R_s) and forced through the load resistance (R_{load}) of the speaker (**1004** in FIG. 7). The HOR feedback resistances (R_{h1} , R_{h2}) sense the voltage across the sense resistance R_s , and such a sensed voltage works against the DAC output current. The closed-loop with such a feedback can force the output current gain of the PWM amplifier **1022** to be adjusted to be $G=1/R_s$. If there is no further adjustment, then the net end-to-end gain in the HOR mode would be $G=R_{load}/R_s$. If R_s and R_{load} are not equal to each other, then the HOR gain will not be equal to the gain in ZOR mode ($G=1$). To address such an effect, an audio amplifier circuit as described herein can configured to include an HOR gain calibration functionality utilizing, for example, adjustment of the sense resistance and a digital gain term to make the gain in HOR mode equal to or approximately equal to the gain in the ZOR mode. Examples related to such a gain calibration functionality are described herein in greater detail.

(87) FIG. 8 shows a block diagram of a digital logic circuit block **1020** that is a more specific example of the digital logic circuit block **1022** of FIG. 5, implemented to operate with the H-bridge driver **1052** of FIG. 7 to provide functionalities including mode switching between HOR and ZOR modes. Such mode switching can be achieved as follows.

(88) It is noted that the audio amplifier circuit as described herein can provide dynamic switching between HOR and ZOR modes. Such mode-switching operation can include switching of the feedback resistance configurations of the resistance network (**1080** in FIG. 8) between the output

nodes (HPP, HPN) to the loop filter block (**1046**). Such mode-switching operation can be achieved in a dynamic manner with low audio artifacts. However, when switching between modes, because of the complex impedance of the speaker driver element and the current-mode operation in the HOR mode, the end-to-end frequency response of the system may change. If such a difference in frequency response is not compensated, the change may result in audible artifacts. To compensate for the change in frequency response, the audio amplifier circuit as described herein can include a digital EQ filter to allow compensation of the frequency response difference between the two modes.

(89) In addition, to minimize the audible artifacts during HOR/ZOR transitions, the audio amplifier circuit as described herein can include a number of features. For example, the resistance network **1080** can be controlled to provide stepped output resistance R_{out} . In another example, a HOR/ZOR EQ block **1142** can be configured to operate with the stepped R_{out} values. In yet another example, a HOR/ZOR state control block **1062** can be provided and configured to control the HOR/ZOR transitions.

(90) It is noted that an abrupt transition in the output resistance R_{out} seen by the speaker driver during HOR/ZOR transitions can cause a sufficiently large phase shift to be audible. To reduce or eliminate such audible artifacts, an amplifier equivalent R_{out} can be made to transition more gradually by moving through a series of R_{out} steps (e.g., six R_{out} steps) during a transition between HOR and ZOR modes. A set of particular R_{out} step values can be selected by selected values of feedback resistances R_h and R_z of the resistance network **1080**. Such R_{out} step values can be selected to produce approximately equal phase artifacts error per step. Further, time duration per step can be programmable over a modest range. Given a non-linear relationship between phase error and step size, R_{out} stepping as described herein can provide a significant impact on the reduction in the audibility of the artifacts.

(91) As described in reference to FIG. 6, an HOR/ZOR EQ block **1142** can be provided as part of the digital audio path **1042**. FIG. 8 shows that such an HOR/ZOR EQ block (also **1142**) can be implemented to operate with the foregoing R_{out} stepping functionality.

(92) Referring to FIG. 8, the HOR/ZOR EQ block **1142** is shown to be driven by the output of the PDM LPF block **1140**. The HOR/ZOR EQ block **1142** can be configured to provide filtering to compensate for the difference between the HOR and ZOR mode frequency responses driving the speaker transducer. The HOR/ZOR EQ block **1142** is shown to include an EQ filter bank, gain compute block and cross-fading functionality.

(93) More particularly, the EQ filter bank is shown to include three parallel filter sections **1200** (Spare BQ), **1202** (Shelving) and **1204** (Bandpass) configured to compensate the frequency response in the speaker transducer. The filter section **1200** (Spare BQ) can be implemented as a low frequency 2nd order (biquad) bandpass filter section (e.g., up to 1 KHz). The filter section **1202** (Shelving) can be implemented as a 4th order finite impulse response (FIR) shelving filter. The filter **1204** (Bandpass) can be implemented as a 2nd order general purpose biquad filter.

(94) Referring to FIG. 8, the EQ block **1142** can be configured to optionally provide filtering to compensate for the difference between the HOR and ZOR mode frequency responses driving the speaker transducer. In addition to the foregoing EQ filter bank and the cross-fading functionality, the EQ block **1142** can also include a gain compute block **1212**. FIG. 9 shows a more specific configuration of such a gain compute block.

(95) Referring to FIGS. 8 and 9, the EQ block **1142** is shown to provide tapered in and out via a crossfading mixing structure of gains G_1 and G_2 (through mixers **1208**, **1210**) and an output adder **1214**.

(96) More particularly, gain G_1 is shown to be associated with mixing of an unfiltered output of the PDM LPF block **1140** with a non-EQ gain signal from the gain compute block **1212** at the mixer **1208**, and gain G_2 is shown to be associated with mixing of a summed filtered signal with an EQ gain signal from the gain compute block **1212** at the mixer **1210**. The foregoing summed filtered

signal is shown to be obtained by outputs of the filter sections **1200**, **1202**, **1204** being added by an adder **1206**. The input to each of the filter sections **1200** (Spare BQ) and the filter section **1202** (Shelving) is shown to be provided from the output of the PDM LPF block **1140**, and the input to the filter section **1204** (Bandpass) is shown to be provided from the input of the PDM LPF block **1140**.

(97) Referring to FIGS. **8** and **9**, the gain value **G1** is associated with a non-filtered signal, and the gain value **G2** is associated with a filtered signal. The gains **G1** and **G2** can be values ranging from 0 to 1, where $G1+G2=1$, and in the example context of Rout having six steps, each of **G1** and **G2** can have six discrete step values of 0, 0.2, 0.4, 0.6, 0.8 and 1. As described herein, such gain values can be stepped with the Rout stepping of the audio amplifier circuit.

(98) Referring to FIG. **9**, the gain compute block **1212** is shown to be configured to generate the non-EQ gain signal (to the mixer **1208** in FIG. **8**) and the EQ gain signal (to the mixer **1210**) based on a number of inputs. For example, the gain compute block **1212** can handle slowly changing gain parameters such as HOR/ZOR gain, user gain, HOR calibration gain, and EQ biquad gain. Based on some or all of such inputs, and depending on the operating mode (ZOR or HOR), the gain compute block **1212** can compute net EQ path gain and non-EQ path gain values and provide such gain values as outputs.

(99) More particularly, the user gain is shown to be processed through a gain ramp **1220** and provided to a mode gain multiplexer **1222** that also receives a HOR mode signal. The mode gain multiplexer **1222** is shown to generate ZOR gain and HOR gain and provide such outputs to a multiplexer **1226**, with the latter being mixed with a HOR calibration gain value by a mixer **1224**. The multiplexer **1226** is shown to provide the non-EQ gain and EQ gain values based on the ZOR gain and HOR gain values, along with an EQ select input. The EQ gain value is shown to result from mixing of the respective output of the multiplexer **1226** with the EQ biquad gain by a mixer **1228**.

(100) Configured in the foregoing manner, the gain compute block **1212** can provide some or all of the following functionalities: provide a gain stage for digital audio gain adjustment, provide an amplifier state initiated gain ramp-up after start of audio, provide an amplifier initiated gain ramp-down before shutdown of audio, provide a gain stage for HOR/ZOR fine gain calibration (e.g., HOR calibration gain) where the fine gain calibration can be applied in all HOR modes (full and partial), provide gain control for a programmable ramp time, and provide a test mode programmable volume gain register to mute gain block with a variable attenuation step size to provide a desired dynamic range.

(101) Configured as described above in reference to FIGS. **8** and **9**, the HOR/ZOR EQ block **1142** can provide some or all of the following functionalities: enabling or bypassing of EQ functionality, compensating for transducer impedance vs frequency with a multi-section filter, switching in and out of the audio path with minimal or reduced audio artifacts, tracking of output resistance Rout, programmable filter coefficients, minimized or reduced latency, and provide EQ filtering active for HOR or ZOR mode and off for the other.

(102) FIG. **8** also shows an HOR/ZOR state control block **1062** that can be a more specific example of the HOR/ZOR state control block **1062** of FIGS. **5** and **6**. In the example of FIG. **8**, the HOR/ZOR state control block **1062** can determine when to switch between HOR and ZOR modes by observing the audio signal path. Such a switching can be controlled so that audible artifacts due to switching are minimized or reduced. In some embodiments, HOR/ZOR switching can be configured to maximize or increase time in the ZOR mode for efficiency, but prioritize HOR mode when audio signal is low in order to minimize or reduce injected interference. The HOR/ZOR state control block **1062** can include an HOR/ZOR Rout mode control block **1190** that utilizes, for example, psychoacoustic principles, to mask or reduce transient artifacts during mode switching operations.

(103) Referring to FIG. **8**, the HOR/ZOR state control block **1062** can provide some or all of the

following functionalities. First, the audio amplifier circuit can be placed in HOR mode when amplitude of the digital audio signal (e.g., obtained from the input side of the PDM LPF **1140**) is low in order to minimize or reduce injected interference. Second, the audio amplifier circuit can be placed in ZOR mode when amplitude of the digital audio signal is high in order to improve efficiency. Third, transition time from ZOR to HOR mode can be made to be within a selected time duration. Fourth, Rout transitions can be made without or reduced audible artifacts. Fifth, HOR/ZOR target mode can be determined by an average audio level. Sixth, HOR/ZOR switching time can be set to be when the transition will be inaudible as determined by a combination of configurable factors such as crest factor, audio signal level being below a threshold value, and time from the last Rout change. Seventh, HOR/ZOR output impedance can have multiple (e.g., 6) steps corresponding to the Rout modes (e.g., Rour0, Rout1, Rout2, Rout3, Rout4, Rout5). Eighth, HOR/ZOR impedance steps can be controlled by moving from current impedance step toward the target impedance step when transitions are determined to be inaudible. Ninth, HOR/ZOR transitions can be programmable to step through discrete output impedances in order to minimize or reduce switching transients (e.g., output impedance step controls being implemented to correspond to output impedance steps of the audio amplifier). Tenth, HOR/ZOR transitions can be independently configured to occur with a minimum or reduced time between switching. Eleventh, HOR/ZOR mode can have a manual override configured by resister access, and such an override can be configured to allow setting of all mode and/or Rout settings.

(104) In the example of FIG. **8**, switching between HOR and ZOR modes can be achieved as follows. Switching between HOR and ZOR modes can include operations of the resistance network **1080**, the HOR/ZOR EQ block **1142**, and the HOR/ZOR state control block **1062**.

(105) Referring to FIG. **8** and the foregoing HOR/ZOR switching functionality, the resistance network **108** can be utilized to provide an Rout stepping functionality as described herein. It is noted that an abrupt transition in Rout seen by the speaker driver during HOR/ZOR transitions can cause a sufficiently large phase shift to be audible. To reduce such audible artifacts, an amplifier equivalent Rout can be made to transition more gradually by moving through a number of Rout steps (e.g., 6 steps) during a transition between HOR and ZOR modes. Such stepped Rout values can be achieved through selection of resistance values for Rh and Rz implemented as, for example, variable resistors. The equivalent Rout steps can be selected to produce approximately equal phase artifact error per step, and the time per step can be programmed over a selected range. Given the non-linear relationship between phase error and step size, Rout stepping functionality can provide a significant impact on the reduction in the audibility of the artifacts during HOR/ZOR transitions.

(106) Referring to FIG. **8** and the foregoing HOR/ZOR switching functionality, the HOR/ZOR EQ block **1142** can be configured to provide EQ filtering functionality including EQ filtering functionality for compensation of the difference in frequency responses between HOR and ZOR modes as described herein. The HOR/ZOR EQ block **1142** can be configured to operate with such EQ filtering functionality during the HOR or ZOR mode. In some embodiments, some or all of the EQ block **1142** can be disabled in the HOR mode to save power, and be enable during the ZOR mode if desired or needed. It is noted that if the foregoing Rout stepping functionality is utilized, the EQ filter response can be synchronized to be gradually stepped with a corresponding change in the Rout.

(107) Referring to FIG. **8** and the foregoing HOR/ZOR switching functionality, the HOR/ZOR state control block **1062** can be configured to decide when to switch between HOR and ZOR modes by monitoring the audio signal path. For example, audio signal path before and after the PDM LPF block **1140** can be monitored.

(108) Referring to FIG. **8**, the node on the input side of the PDM LPF block **1140** is shown to be coupled to a decimation circuit **1180** configured decimate the sampled signal by N (e.g., N=128). The decimated signal is then shown to be provided to a block **1182** for low audio detection, a block **1184** for generating a target mode, and a block **1186** for selecting an Rout step. Outputs of the

blocks **1182**, **1184**, **1186** are shown to be provided to a HOR/ZOR Rout mode control block **1190**. Referring to FIG. **8**, the node on the output side of the PDM LPF block **1140** is shown to be coupled to an audio switch block **1188** that provides its output to the HOR/ZOR Rout mode control block **1190**.

(109) Configured in the foregoing manner, the HOR/ZOR state control block **1062** can determine when to switch between HOR and ZOR modes. Such determination can be based on some or all of a number of conditions. For example, an average of the sampled input signal can be obtained and compared to a threshold value. More particularly, a determination of whether the average of the sampled signal is greater than a HOR-to-ZOR threshold value can be made when in HOR mode, or whether the average of the sampled signal is less than a ZOR-to-HOR threshold value can be made when in ZOR mode. In another example, a determination of whether the input signal has a very low level can be made. In yet another example, the input signal can be passed through a high-pass filter, and a peak in such a filtered signal can be compared to a product of an average of the filtered signal and a crest factor. If the peak value is greater than the product, and if the input signal is at or near a zero crossing, a determination can be made to perform a mode-switching operation.

(110) It is noted that the foregoing HOR-to-ZOR threshold value and ZOR-to-HOR threshold value may or may not be different.

(111) For a HOR-to-ZOR transition, determination can be made as to whether the input signal level has crossed a respective threshold level, and whether a favorable transition condition (e.g., high crest factor, higher frequency masking event and a subsequent zero crossing). If so, the HOR-to-ZOR transition can be made to proceed. For a ZOR-to-HOR transition, determination can be made as to whether the input signal level has crossed a respective threshold level or has become sufficiently small, and whether a favorable transition condition (e.g., high crest factor, higher frequency masking event and a subsequent zero crossing). If so, the ZOR-to-HOR transition can be made to proceed. The foregoing transition techniques allow the audio amplifier to operate to deliver full output power in either HOR or ZOR modes, thus enabling the audio amplifier to remain in either mode until a favorable transition condition is present to thereby avoid audible artifacts.

(112) As described herein, HOR mode gain is determined differently than ZOR mode gain. However, it is desirable to operate an audio amplifier circuit as described herein so that a net gain in the HOR mode is equivalent to a net gain in the ZOR mode. In some embodiments, HOR gain can be adjusted so to be equivalent to ZOR gain. In various examples described herein, such a ZOR gain can be considered to have a gain $G=1$; thus, HOR gain can be calibrated to also provide a gain $G=1$. Such a HOR gain calibration can be achieved as follows.

(113) FIG. **10** shows various functional blocks of the audio amplifier circuit **1000** of FIG. **5**, where such blocks can form an HOR gain calibration sub-system. More particularly, such a sub-system can include a gain calibration block **1070**, a gain calibration ADC **1084**, an Rout controller **1064**, a resistance network **1080**, and an HOR/ZOR EQ block **1142**. FIG. **11** shows a more detailed example of the gain calibration block **1070** of FIG. **10**.

(114) Referring to FIGS. **10** and **11**, the gain calibration block **1070** can be implemented as a digital block that includes a gain calibration processor **1230** and a gain calibration controller **1238**. The gain calibration processor **1230** can be configured to perform gain estimation computation for HOR mode, and based on such gain estimation, desired sense resistance (R_s in FIGS. **7** and **8**) value and digital gain correction value can be determined. The gain calibration controller **1238** can handle overall management of the gain calibration sub-system, including determining when to calibrate and/or determining conditions for a valid calibration.

(115) Referring to FIGS. **10** and **11**, the gain calibration ADC **1084** can be implemented as an analog block that digitizes the differential amplifier output HPP-HPN during a calibration cycle to use by the foregoing gain calibration processor **1230**. The gain calibration ADC **1084** can be configured (e.g., with delta-sigma modulation) to handle large audio signals at the output (HPP, HPN) and have a sufficiently large dynamic range to detect the relatively lower amplitude

calibration tone signal.

(116) Referring to FIG. 11, the gain calibration processor **1230** is shown to include a calibration tone generator **1232** configured to generate an ultrasonic calibration tone (Cal. tone) which is mixed into the digital audio stream with the mixer **1144** before the signal limiter block **1146**. The tone generator **1232** can also be configured to provide programmable frequency and amplitude functionalities. In some embodiments, the tone generator **1232** can produce a sinusoidal output modulated with, for example, a 2nd order cascaded integrator-comb (CIC) filtered pulse to prevent audible energy during turning on and off of the tone generator **1232**.

(117) Referring to FIGS. 10 and 11, the Rout controller **1064** can be implemented as a digital block that performs computations to determine and control the sense resistance (R_s) setting in the analog resistance network **1080**. The Rout controller **1064** can also perform computations to determine a digital fine gain value that is provided to the HOR/ZOR EQ block **1142** for use in a fine gain adjustment.

(118) Referring to FIGS. 10 and 8, the resistance network **1080** can be implemented as an analog block that includes a programmable sense resistance (R_s) network, and programmable ZOR and HOR Rout mode feedback resistances (R_z and R_h). Such a resistance network is shown to be controlled by the Rout controller **1064**.

(119) Referring to FIGS. 10 and 11, and as described above, the HOR/ZOR EQ block **1142** and its gain compute block (**1212** in FIGS. 8 and 9) can provide a gain multiplier functionality where HOR digital calibration gain is applied. Such an application of the HOR digital calibration gain can be based on a control signal (HOR Cal fine gain) provided by the Rout controller **1064**.

(120) Configured in the foregoing manner, the gain calibration processor **1230** can compute a gain mismatch between HOR and ZOR modes by introducing a calibration tone (e.g., an ultrasonic tone at ~25 KHz) into the digital audio stream at the mixer **1144** before the signal limiter block **1146**. The digital audio stream with the calibration tone mixed therein is passed through the signal limiter block **1146**. At the output of the signal limiter block **1146**, the digital audio stream is routed through path **1043** to the PWM controller (**1050** in FIG. 8) to be processed and amplified by the H-bridge driver **1052** to provide an output at HPP and HPN. Also at the output of the signal limiter block **1146**, the digital audio stream is obtained for the gain calibration processor **1230** as a reference signal. Thus, the magnitude and phase of the output signal at the output (HPP, HPN) can be compared relative to the magnitude and phase of the reference signal.

(121) To achieve the foregoing comparison of the output signal (analog signal) with the reference signal (digital signal), the gain calibration ADC **1084** can sample the output voltage across the load (HPP-HPN) and provide a delta-sigma ADC output to the gain calibration processor **1230**.

Measurement and computation of the HOR and ZOR gains can be performed by the gain calibration processor **1230** (e.g., with a gain compute block **1236**) utilizing an estimation algorithm where the reference signal X and the digitized output signal Y are downconverted with a tone at the same frequency as the calibration tone (e.g., 25 KHz). The downconverted signals X and Y can be filtered (e.g., single-bin fast Fourier transform (FFT) with a discrete Fourier transform (DFT) block **1234**) to provide respective complex downconverted values x and y. A ratio of the two complex downconverted values can be obtained, where $\text{Ratio} = y/x$. It is noted that complex values can be utilized so that the load inductance does not significantly affect the gain calculation; however, the real component of the Ratio is utilized.

(122) The foregoing $\text{Ratio} = y/x$ is an expression of a transfer function gain from the digital input to the amplifier output, and can be designed to be tolerant to out-of-band interference. Ratio can be computed for each of the HOR and ZOR modes, such that $\text{Ratio}(\text{HOR}) = y/x$ in the HOR mode, and $\text{Ratio}(\text{ZOR}) = y/x$ in the ZOR mode.

(123) The calibration processor **1230** can then compute another ratio $\text{Relative_HOR_gain} = \text{Ratio}(\text{HOR}) / \text{Ratio}(\text{ZOR})$ which is representative of the HOR gain relative to the ZOR gain. Ideally, this ratio Relative_HOR_gain has a value of 1.

(124) Referring to FIG. 11, the computed Relative_HOR_gain value is shown to be provided to the Rout control block **1064** by the gain calibration controller **1238**. Based on such a computed Relative_HOR_gain value, the Rout control block **1064** can determine an adjustment to the sense resistance (Rs) and a fine gain (HOR Cal fine gain in FIG. 10) control signal to be applied to the digital path by the gain compute block **1212** of the HOR/ZOR EQ block **1142**.

(125) More particularly, the Rout control block **1064** can obtain the Relative_HOR_gain value and perform computations to determine how to change the sense resistance (Rs) relative to its present setting. Based on the change it makes to the sense resistance (Rs) setting, the Rout control block **1064** can compute the digital fine gain adjustment needed to make the Relative_HOR_gain value to be 1.

(126) As described herein, a digital signal is provided to a PWM controller as a feed-forward signal; and the PWM controller generates a control signal that drives an appropriate portion of an H-bridge driver. The digital signal is also converted into an analog signal, and such an analog signal is utilized as a reference signal that is compared to a feedback signal from the output of the H-bridge driver. FIG. 7 includes examples related to processing of such feed-forward and analog signals. More particularly, the foregoing analog signal can be generated by a digital-to-analog converter (DAC) **1044** based on a digital signal provided to it from a dynamic element matching (DEM) block **1150**. The DEM block **1150** is shown to be a part of a digital logic circuit **1020**, and more particularly, a digital audio path (**1042** in FIG. 6) of the digital logic circuit **1020**.

(127) Referring to FIG. 6, the feed-forward digital signal is shown to be provided to the PWM controller through a path **1043**, from a node upstream of a delta sigma modulator (DSM) **1148**. A digital signal is also shown to be provided to the DSM **1148**, and a digital output of the DSM **1148** is shown to be provided to a DEM block **1150**. A digital output of the DEM block **1150** is shown to be provide to a DAC **1144** through a path indicated as **1045**.

(128) Configured in the foregoing manner, the DSM block **1148** can be configured to re-modulate a higher-bit input signal (e.g., 24-bit signal), through delta-sigma modulation, into a lower-bit signal (e.g., 9-bit signal) that is appropriate for the DEM block **1150**. Accordingly, the example 9-bit DSM output signal drives the DEM block **1150** which can be implemented as a digital block configured to, for example, randomize a pattern of 512-bit cell drive to the DAC **1044** in a manner to linearize the DAC's response for use as a multi-bit delta-sigma DAC. Such a configuration can provide a desirable reference audio signal for the closed-loop circuit described herein.

(129) FIG. 12A depicts a digital-to-analog converter (DAC) **10** having N-bit resolution, where N is a positive integer. Accordingly, the DAC **10** can be referred to as an N-bit DAC. Such a DAC includes an array **12** of N bit cells **14a**, **14b**, **14c**, etc., and each bit cell is provided with a voltage V and a control signal **b0**, **b1**, **b2**, etc. Each cell includes a set resistance and a switch, under the control of the respective control signal, that allows the voltage node (V) to be connected to a common node **16** through the resistance. Depending on the type of DAC, there may be a resistance between neighboring bit cells.

(130) Configured in the foregoing manner, each bit of a digital signal presents either no voltage or a resistance-reduced voltage from the corresponding bit cell to an adder circuit **18** to generate an analog output voltage value Vout representative of the digital signal.

(131) For the purpose of description, FIG. 12B depicts the DAC **10** of FIG. 12A in a block form. More particularly, an N-bit cell array **12** is assumed to include N bit cells as described in reference to FIG. 12A, and N-bit data being applied to the N-bit cell array **12** is assumed to include control signals for the N bit cells. Similarly, a voltage V being provided to the N-bit cell array **12** is assumed to be distributed to the N bit cells.

(132) For example, FIG. 13A shows an example where a DAC **10** is a 3-bit DAC, such that an array **12** includes bit cells **14a**, **14b**, **14c**. Each bit cell is shown to be provided with a voltage V and a respective control signal (**b0**, **b1**, **b2**). An output of each bit cell is shown to be coupled to a common node **16** which is coupled to an adder circuit **18** to generate an analog output signal Vout.

(133) In the block form of FIG. 13B, the array 12 is indicated as a 3-bit cell array, and the control signals for the three bit cells are indicated as 3-bit data.

(134) It is noted that the DACs of FIGS. 12 and 13 are examples of fixed bit DACs where the number of bit cells is fixed for each DAC. Accordingly, all of the bit cells in a given DAC are operable.

(135) FIG. 14 shows that in some embodiments, a DAC 100 can include a variable bit cell array size. In some embodiments, such a DAC having variable bit cell array size can be utilized as the DAC 1044 of FIGS. 5 to 8.

(136) In electronic applications such as audio signal processing applications, low power and space constrained DACs typically use small (low resolution) bit cell arrays. DACs with large arrays bit cells can be implemented to obtain high resolution; however, such implementations result in challenges such as larger size requirement, higher noise, and increased quiescent power consumption.

(137) In some embodiments, a DAC architecture having a telescoping functionality can be implemented to allow dynamic adjustment of the DAC's cell population being used. In such an architecture, a large DAC array can be enabled and utilized without the permanent presence of the foregoing challenges.

(138) For the purpose of description, the foregoing DAC architecture having a telescoping functionality also may be referred to herein as a dynamic element matching (DEM) DAC or a telescoping DEM DAC.

(139) FIG. 14 depicts a DAC 100 having an array 102 of N bit cells, where the quantity N is a positive integer that can vary between a lower limit $N_{\text{sub.low}}$ to an upper limit $N_{\text{sub.high}}$ so as to provide a telescoping functionality indicated as an arrow 106. For a given value of N, N cells of the array 102 can be operated by application of a voltage V and an N-bit data, so as to provide (arrow 104) respective voltages to an adder 108 to generate an analog output signal.

(140) In some embodiments, various functionalities, including the telescoping functionality, can be controlled and/or supported by a controller 110. Examples of such control functionalities are described herein in greater detail. In some embodiments, such a controller can be a part of the DAC 100, be external to the DAC 100, or some combination thereof.

(141) It is noted that operationally, a DAC having a large fixed array can also suffer from band artifacts when too few cells are being activated on average in quiescent conditions compared to the total cell population. In some embodiments, a DAC architecture (such as the DAC 100 of FIG. 14) having a telescoping functionality can provide dynamic array population control that pushes the foregoing band artifacts out of band by utilizing only a small set of cells at low levels associated with quiescent conditions.

(142) It is also noted that a high resolution DAC array typically uses more power to overcome various implementation challenges. On the other hand, a low power design typically has low resolution. Thus, pushing DEM artifacts out of band can be achieved by either adding out of band signal to the DAC codes or biasing some number of cells to be always on; however, each of such approaches increases power consumption.

(143) In some embodiments, the DAC architecture 100 of FIG. 14 can be configured to use a dynamic subset of the array 102 of DAC bit cells. For example, at low output levels, only a small subsection of the array 102 are utilized. In such a configuration, most of the analog circuitry can be shut off, and most of the digital circuitry can be static to reduce current.

(144) When a signal requires more cells than are currently available, additional cells can be added to the active population along with their corresponding digital control circuitry. This allows use of a high precision DAC array while maintaining low quiescent current and eliminating or reducing DEM artifacts in the audio band.

(145) Increasing or decreasing the DAC cell population also changes the gain of the DAC. In some embodiments, either or both of time and level hysteresis can be utilized to avoid or reduce audible

artifacts from dynamically changing the active cell population.

(146) In some embodiments, active cell population is not increased until a telescope threshold is exceeded for a programmable number of cycles, to avoid population chatter due to noise occasionally exceeding the threshold. In some embodiments, active cell population is not decreased until a signal peak remains below a telescope threshold for a programmable time to guarantee that a constant AC signal will not change the population size cycle to cycle and produce total harmonic distortion.

(147) In some embodiments, the DAC architecture **100** of FIG. **14** can be configured such that the DEM functionality is accomplished with a variable length barrel shifter that drives a scrambling network. Barrel shifter operation can enforce an order on bit cell usage resulting in bit cell mismatch noise to be pushed out of band. The scrambling network can occasionally randomize the cell order to reduce tones and other artifacts caused by the barrel repeat rate.

(148) In some embodiments, the scrambling network can be implemented as a butterfly or Benes network fed by a linear-feedback shift register (LFSR). An output of the LFSR and its compliment can be distributed to randomize all of the switches in the scramble network. The LFSR can be clocked at a programmable time interval to set the randomization period.

(149) Referring to FIGS. **14** and **5** to **8**, it is noted that in some embodiments, the DAC (**100** or **1044**) can be implemented as an 8-bit plus sign delta-sigma modulated DAC to produce a reference audio signal for the loop circuit of the PWM amplifier (**1022** in FIG. **7**). Such a DAC preferably operates with very low noise and support a high dynamic range, with low distortion to meet a high performance THD specification, and with low power consumption to meet a low quiescent power specification. In some embodiments, a 512-unit cell having telescoping architecture as described herein can be utilized.

(150) In some embodiments, a DAC having a telescoping set of unit bit cells as described herein can be configured to operate in one of a number of modes. For example, four modes can be implemented, where a first mode utilizes the least amount of the unit bit cells, and second to fourth modes utilize successively greater numbers of unit bit cells. In the lowest telescope mode (first mode), the fewest current bit cells are active; and in the highest telescope mode (fourth mode in the 4-mode example), the most current bit cells are active. In some embodiments, the amplitude of the input signal and the audio operating mode can control which telescope mode the DAC is in at any given time.

(151) It is noted that for a given number of current bit cells in a DAC (e.g., 512 total bit cells for an 8-bit plus sign input), the bit cells match with each other to only a certain level. Thus, if such bit cells are addressed sequentially, distortion can be introduced to thereby limit performance.

(152) In some embodiments, and as described herein, a digital dynamic element matching (DEM) algorithm in a digital circuit (DEM block **1150** in FIGS. **5** to **7**) can be configured to effectively shuffle and/or randomize the usage of bit cells when decoding the input signal (e.g., 8-bit signal) to mapping of the bit cells (e.g., among 512 bit cells). Such a functionality can provide significant improvement in distortion performance at the DAC output.

(153) In some embodiments, a bias DEM block can be implemented to provide analogous shuffling function on the reference current cell side to improve, for example 1/frequency noise performance and matching. In the context of the example four telescope modes of the DAC, the bias DEM block can include three telescope modes, and an operating mode can be selected from such three telescope modes based on, for example, signal-to-noise ratio (SNR) and input signal level.

(154) In some embodiments, the DAC of FIGS. **5** to **7** and **14** can utilize a local DAC reference voltage to drive the current source bit cells. Such reference can be configured for lower power and noise; however, may include a larger temperature dependence. Thus, in some embodiments, the local DAC reference voltage can be calibrated (e.g., periodically) against a trimmed main voltage reference to ensure that gain accuracy is maintained over a range of temperature.

(155) In some embodiments, the DAC's absolute gain will have been trimmed in production

testing, and such a trimmed gain can be utilized as a comparison for calibration. Such a calibration can occur in background while the audio amplifier circuit is operating in audio mode. If there is an updated needed or desired to the reference setting, an adjustment can be made and applied during, for example, the next audio signal zero crossing to minimize or reduce any artifacts.

(156) The present disclosure describes various features, no single one of which is solely responsible for the benefits described herein. It will be understood that various features described herein may be combined, modified, or omitted, as would be apparent to one of ordinary skill. Other combinations and sub-combinations than those specifically described herein will be apparent to one of ordinary skill, and are intended to form a part of this disclosure. Various methods are described herein in connection with various flowchart steps and/or phases. It will be understood that in many cases, certain steps and/or phases may be combined together such that multiple steps and/or phases shown in the flowcharts can be performed as a single step and/or phase. Also, certain steps and/or phases can be broken into additional sub-components to be performed separately. In some instances, the order of the steps and/or phases can be rearranged and certain steps and/or phases may be omitted entirely. Also, the methods described herein are to be understood to be open-ended, such that additional steps and/or phases to those shown and described herein can also be performed.

(157) Some aspects of the systems and methods described herein can advantageously be implemented using, for example, computer software, hardware, firmware, or any combination of computer software, hardware, and firmware. Computer software can comprise computer executable code stored in a computer readable medium (e.g., non-transitory computer readable medium) that, when executed, performs the functions described herein. In some embodiments, computer-executable code is executed by one or more general purpose computer processors. A skilled artisan will appreciate, in light of this disclosure, that any feature or function that can be implemented using software to be executed on a general purpose computer can also be implemented using a different combination of hardware, software, or firmware. For example, such a module can be implemented completely in hardware using a combination of integrated circuits. Alternatively or additionally, such a feature or function can be implemented completely or partially using specialized computers designed to perform the particular functions described herein rather than by general purpose computers.

(158) Multiple distributed computing devices can be substituted for any one computing device described herein. In such distributed embodiments, the functions of the one computing device are distributed (e.g., over a network) such that some functions are performed on each of the distributed computing devices.

(159) Some embodiments may be described with reference to equations, algorithms, and/or flowchart illustrations. These methods may be implemented using computer program instructions executable on one or more computers. These methods may also be implemented as computer program products either separately, or as a component of an apparatus or system. In this regard, each equation, algorithm, block, or step of a flowchart, and combinations thereof, may be implemented by hardware, firmware, and/or software including one or more computer program instructions embodied in computer-readable program code logic. As will be appreciated, any such computer program instructions may be loaded onto one or more computers, including without limitation a general purpose computer or special purpose computer, or other programmable processing apparatus to produce a machine, such that the computer program instructions which execute on the computer(s) or other programmable processing device(s) implement the functions specified in the equations, algorithms, and/or flowcharts. It will also be understood that each equation, algorithm, and/or block in flowchart illustrations, and combinations thereof, may be implemented by special purpose hardware-based computer systems which perform the specified functions or steps, or combinations of special purpose hardware and computer-readable program code logic means.

(160) Furthermore, computer program instructions, such as embodied in computer-readable

program code logic, may also be stored in a computer readable memory (e.g., a non-transitory computer readable medium) that can direct one or more computers or other programmable processing devices to function in a particular manner, such that the instructions stored in the computer-readable memory implement the function(s) specified in the block(s) of the flowchart(s). The computer program instructions may also be loaded onto one or more computers or other programmable computing devices to cause a series of operational steps to be performed on the one or more computers or other programmable computing devices to produce a computer-implemented process such that the instructions which execute on the computer or other programmable processing apparatus provide steps for implementing the functions specified in the equation(s), algorithm(s), and/or block(s) of the flowchart(s).

(161) Some or all of the methods and tasks described herein may be performed and fully automated by a computer system. The computer system may, in some cases, include multiple distinct computers or computing devices (e.g., physical servers, workstations, storage arrays, etc.) that communicate and interoperate over a network to perform the described functions. Each such computing device typically includes a processor (or multiple processors) that executes program instructions or modules stored in a memory or other non-transitory computer-readable storage medium or device. The various functions disclosed herein may be embodied in such program instructions, although some or all of the disclosed functions may alternatively be implemented in application-specific circuitry (e.g., ASICs or FPGAs) of the computer system. Where the computer system includes multiple computing devices, these devices may, but need not, be co-located. The results of the disclosed methods and tasks may be persistently stored by transforming physical storage devices, such as solid state memory chips and/or magnetic disks, into a different state.

(162) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list. The word “exemplary” is used exclusively herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

(163) The disclosure is not intended to be limited to the implementations shown herein. Various modifications to the implementations described in this disclosure may be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. The teachings of the invention provided herein can be applied to other methods and systems, and are not limited to the methods and systems described above, and elements and acts of the various embodiments described above can be combined to provide further embodiments. Accordingly, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A digital-to-analog converter (DAC) architecture comprising: an array having a total number of bit cells; and a control system configured to activate a selected number of the total number of bit cells and to deactivate the remaining bit cells, the selected number being variable, such that the array consumes a quiescent current that depends on the selected number, the control system further configured to change the selected number when a signal condition exceeds a threshold duration, the DAC architecture being configured to convert a digital signal stream into a respective analog audio signal stream, and the control system including a variable length shifter implemented to perform a shifting operation among the selected number of active bit cells.
2. The DAC architecture of claim 1 wherein the control system selects a low value for the selected number when a low resolution is sufficient, such that the array consumes a low amount of quiescent current when the selected number is low.
3. The DAC architecture of claim 1 wherein the control system selects a high value for the selected number when a high resolution is desired, such that the array consumes a high amount of quiescent current when the selected number is high.
4. The DAC architecture of claim 1 wherein the control system further includes a scrambling network implemented to be driven by the variable length shifter.
5. The DAC architecture of claim 4 wherein the scrambling network is implemented as a butterfly or Benes network fed by a linear-feedback shift register.
6. The DAC architecture of claim 1 wherein the control system is configured such that the threshold duration includes a first threshold duration for increasing the selected number, and a second threshold duration for decreasing the selected number.
7. The DAC architecture of claim 6 wherein the first threshold duration is approximately equal to the second threshold duration.
8. The DAC architecture of claim 6 wherein the first threshold duration is different than the second threshold duration.
9. The DAC architecture of claim 6 wherein the first threshold duration is selected to avoid an increase in the selected number due to an occasional noise.
10. The DAC architecture of claim 6 wherein the second threshold duration is selected based on the signal condition remaining below a threshold value for the second threshold duration.
11. The DAC architecture of claim 10 wherein the second threshold duration is selected to avoid a decrease in the selected number due to a constant AC signal.
12. The DAC architecture of claim 1 wherein the control system is configured to utilize either or both of time and level hysteresis to avoid or reduce an audible artifact resulting from a gain change associated with a dynamic change in the number of active bit cells.
13. The DAC architecture of claim 1 wherein the variable length shifter includes a barrel shifter configured to perform a barrel shifting operation as the shifting operation.
14. A method for converting a digital signal to an analog signal in a digital-to-analog converter (DAC) architecture, the method comprising: providing an array with a total number of bit cells; and controlling the array to activate a selected number of the total number of bit cells and to deactivate the remaining bit cells, the selected number being variable, such that the array consumes a quiescent current that depends on the selected number, the controlling further including changing the selected number when a signal condition exceeds a threshold duration, the controlling further including a variable length shifting operation among the selected number of active bit cells, such that the DAC architecture converts a digital signal stream into a respective analog audio signal stream.
15. A wireless device comprising: an antenna for receiving a wireless signal; a driver configured to convert a digital signal representative of the received wireless signal into an analog signal with a

digital-to-analog converter (DAC) system, the DAC system including an array having a total number of bit cells, the DAC system further including a control system configured to activate a selected number of the total number of bit cells and to deactivate the remaining bit cells, the selected number being variable, such that the array consumes a quiescent current that depends on the selected number, the control system further configured to change the selected number when a signal condition exceeds a threshold duration, the control system including a variable length shifter implemented to perform a shifting operation among the selected number of active bit cells; and a speaker in communication with the driver and configured to generate sound waves based on the analog signal.

16. The wireless device of claim 14 wherein the wireless device is a wireless headphone or a wireless earphone.

17. The wireless device of claim 15 wherein the variable length shifter includes a barrel shifter configured to perform a barrel shifting operation as the shifting operation.
